Sales Performance and Customer Insights

PROJECT OVERVIEW

This project analyzed four e-commerce datasets: Customers, Products, Orders, and Order Items to understand sales performance and customer purchasing behavior from January 2024 to January 2026. Python, Pandas, Matplotlib, and Streamlit were used for data cleaning, analysis, and visualization.

OVERALL SALES PERFORMANCE

- **Total calculated revenue:** ₱9,237,598.93
- **Unique orders:** 14,701
- **Average order value:** ₱628.37
- **Consolidated transaction records:** 34,689

KEY BUSINESS INSIGHTS

1. **Product Category Performance**

   Electronics generated the highest category revenue at ₱1,282,129.54, followed by Toys at ₱1,255,350.95. These categories are important contributors to overall calculated revenue.

2. **Individual Product Performance**

   KidJoy Toys 607 was the highest-revenue individual product, generating ₱40,445.79.

3. **Geographical Sales Performance**

   Olongapo recorded the highest city-level revenue at ₱591,334.47, followed by Iligan and San Fernando.

4. **Customer Purchasing Behavior**

   Repeat customers generated ₱8,742,510.71, representing 94.64% of total calculated revenue. New customers contributed ₱495,088.23, or 5.36%.

5. **Monthly Revenue Trends**

   October 2024 recorded the highest monthly revenue at ₱405,778.61, while February 2025 recorded the lowest at ₱336,527.87.

RECOMMENDATIONS

- Strengthen customer retention through loyalty programs and repeat-purchase incentives.
- Improve new customer acquisition through targeted marketing and referral programs.
- Prioritize inventory planning for high-performing categories and products.
- Develop location-specific marketing strategies for high-performing cities.
- Monitor monthly revenue and key performance indicators regularly.
- Improve revenue reporting by distinguishing transactions according to order status.

CONCLUSION

The analysis demonstrates how data analytics and visualization can transform raw business data into meaningful insights. The findings can support better marketing strategies, customer engagement, inventory planning, and business decision-making.